

PRODUCT DELIVERY MANAGEMENT SYSTEM:

INPUTS:

DEPT.	NO. OF EMPS.	DURATION	DELIVERY DATE	NOTICE PERIOD EMP. NOS	DELAY EST.
DEVELOPMENT	30	3 MONTHS	26 DEC 2026	5	20 DAYS
TESTING	45	3 MONTHS	26 MAR 2027	3	15 DAYS
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Problem statement: Based on the number of employees available in each department and their notice periods, predict the project delivery delay in days.
Stage 1 – Machine Learning: The inputs are employee/department information, notice periods, project/module requirements, etc., and the target is the expected delay in days.
Stage 2 – Regression: Because the target is a continuous numerical value such as 5 days, 12.5 days, or 20 days, this is a regression problem, not classification.

Solution Overview: Since the problem statement is on number of employees and their notice periods, the expected output is how much delay in days/time will happen for a project delivery. Since the input is number of employees per department to work for a module, we can choose Stage-1 as Machine Learning, since the input requirements are clear, we have to populate the output as estimated time/no. of days/date as a prediction. so we can pick Stage-2 as supervised learning to fill the outputs as the estimated delay in the form of labels (Actual Delay). since we are going to populate continuous numeric values its falls on Regression not a classification.

Employees	Notice Period	Required Employees	Actual Delay
10	30 days	8	5 days
6	60 days	8	20 days
12	15 days	10	2 days

we have labelled data (Actual Delay), so we can directly use supervised regression.

Input:
ML employees = 5
Required employees = 8
Average notice = 30 days
Project duration = 60 days

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Regression Model

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Predicted delay = 18 days